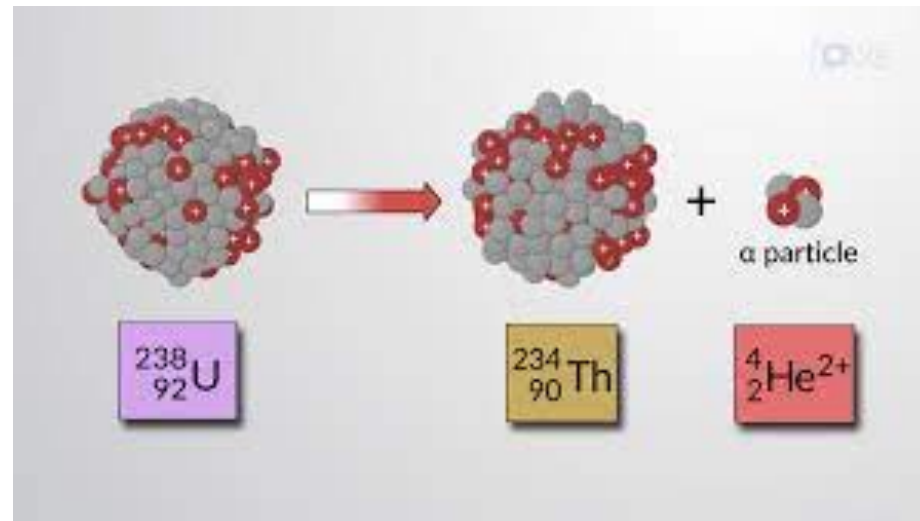


NUCLEAR CHEMISTRY



Nuclear Chemistry

A nuclear reaction is different from a chemical reaction. In a chemical reaction, atoms of the reactants combine by a rearrangement of extranuclear electrons but the nuclei of the atoms remain unchanged. In a nuclear reaction, on the other hand, it is the nucleus of the atom which is involved. The number of protons or neutrons in the nucleus changes to form a new element. **A study of the nuclear changes in atoms is termed Nuclear Chemistry.**



RADIOACTIVITY

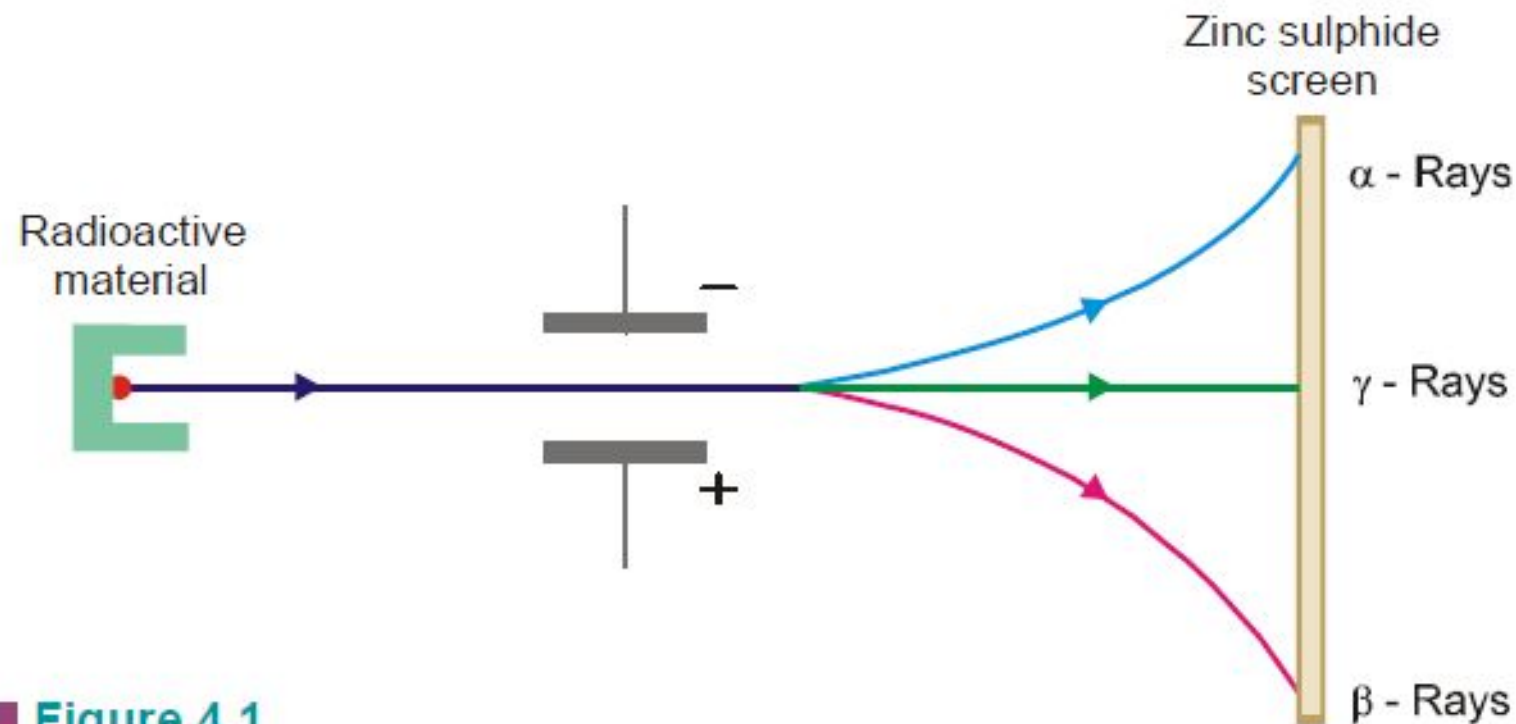
A number of elements such as uranium and radium are unstable. Their atomic nucleus breaks of its own accord to form a smaller atomic nucleus of another element. The protons and neutrons in the unstable nucleus regroup to give the new nucleus.

This causes the release of excess particles and energy from the original nucleus, which we call **radiation**. The elements whose atomic nucleus emits radiation are said to be **radioactive**.

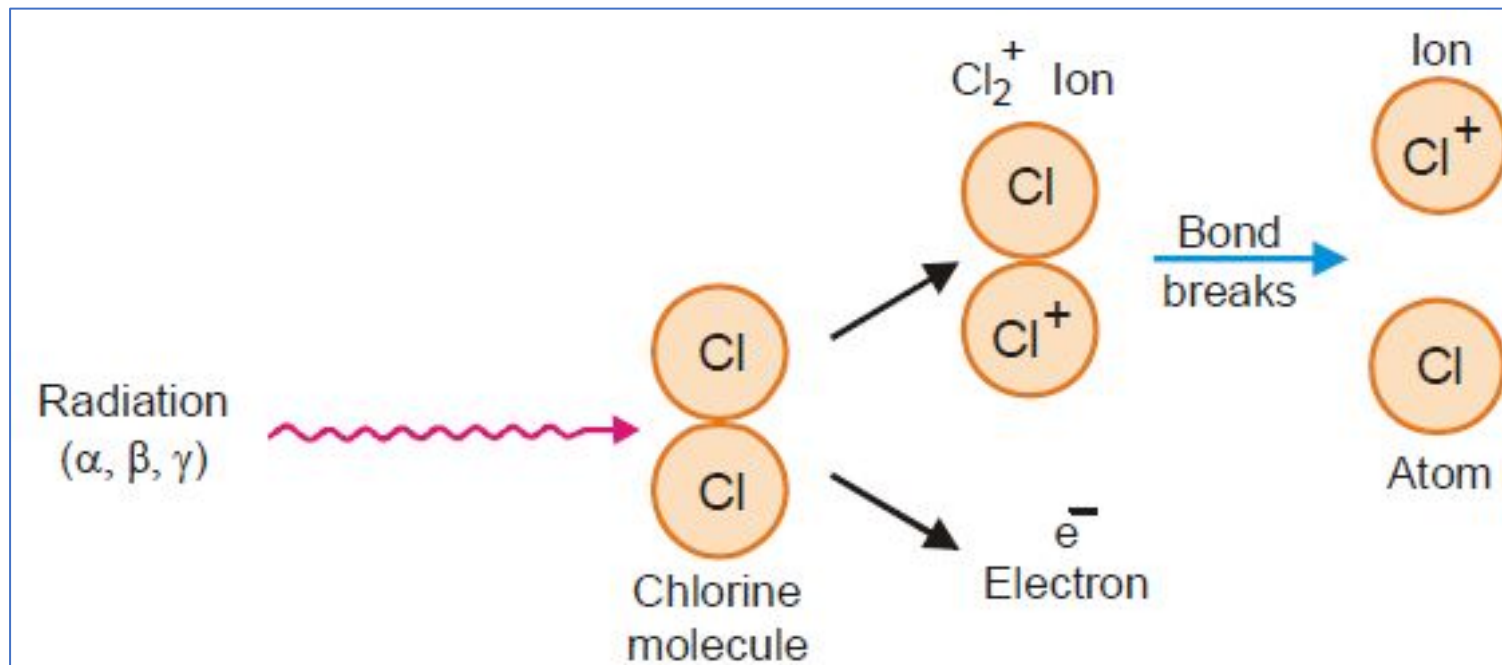
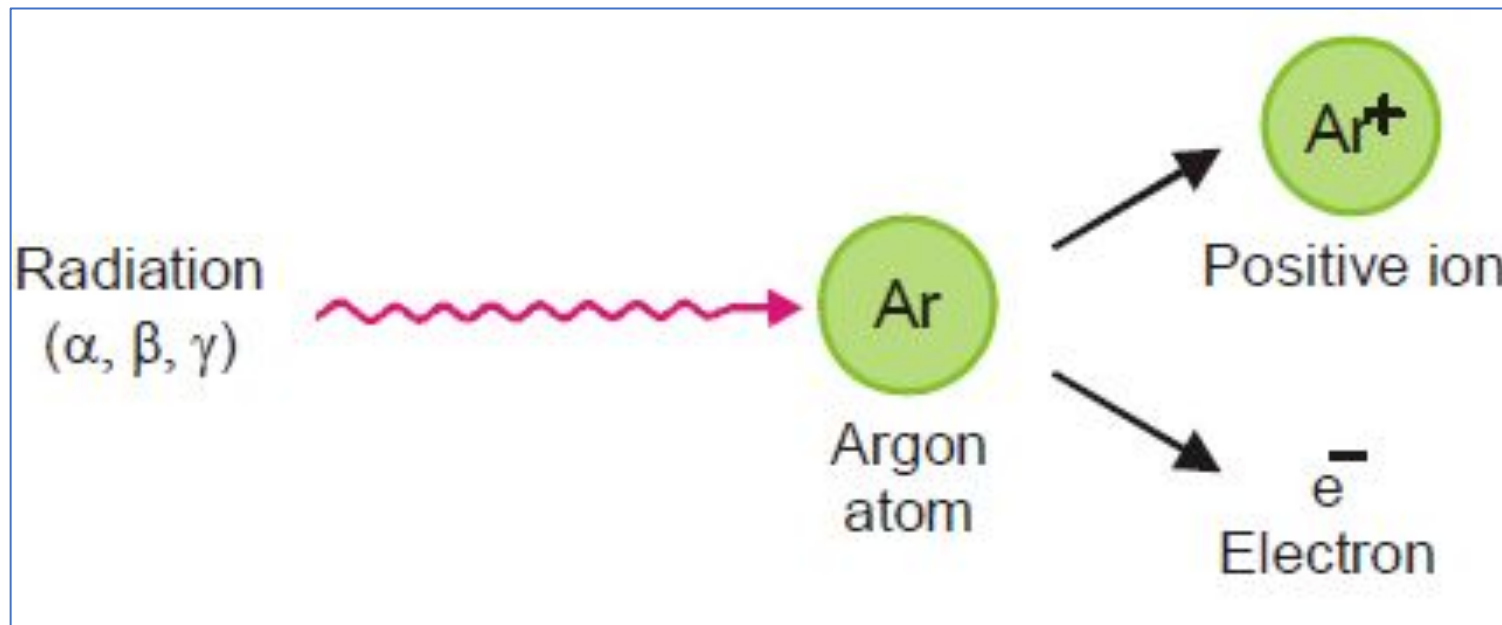
The spontaneous breaking down of the unstable atoms is termed **radioactive disintegration or radioactive decay**. The **disintegration or decay of unstable atoms accompanied by emission of radiation is called Radioactivity**.

TYPES OF RADIATIONS

The radioactive radiations are of three types. These were sorted out by Rutherford (1902) by passing them between two oppositely charged plates (Fig. 4.1). The one bending towards the negative plate carried positive charge and were named α (**alpha**) rays. Those bending towards the positive plate and carrying negative charge were called β (**beta**) rays. The third type of radiation, being uncharged, passed straight through the electric field and were named γ (**gamma**) rays. α , β and γ rays could be easily detected as they cause luminescence on the zinc sulphide screen placed in their path.



■ **Figure 4.1**
Detection of α , β and γ -Rays.



ALPHA RAYS

(1) **Nature.** They consist of streams of α -particles. By measurement of their e/m , Rutherford showed that they have a mass of 4 amu and charge of +2. They are helium nuclei and may be represented as ${}^4_2\alpha$ or ${}^4_2\text{He}$.

(2) **Velocity.** α -particles are ejected from radioactive nuclei with very high velocity, about one-tenth that of light.

(3) **Penetrating power.** Because of their charge and relatively large size, α -particles have very little power of penetration through matter. They are stopped by a sheet of paper, 0.01 mm thick aluminium foil or a few centimetres of air.

(4) **Ionisation.** They cause intense ionisation of a gas through which they pass. On account of their high velocity and attraction for electrons, α -particles break away electrons from gas molecules and convert them to positive ions.

BETA RAYS

(1) **Nature.** They are streams of β -particles emitted by the nucleus. From their deflection electric and magnetic fields, Becquerel showed that β -particles are identical with electrons. They have very small mass (1/1827 amu) and charge of -1 . A β -particle is symbolized as ${}^0_{-1}\beta$ or ${}^0_{-1}e$.

(2) **Velocity.** They travel about 10 times faster than α -particles. Their velocity is about the same as of light.

(3) **Penetrating power.** β -Particles are 100 times more penetrating in comparison to α -particles. This is so because they have higher velocity and negligible mass. β -particles can be stopped by about 1 cm thick sheet of aluminium or 1 m of air.

BETA RAYS

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(3) **Penetrating power.** **β -Particles are 100 times more penetrating in comparison to α -particles**. This is so because they have higher velocity and negligible mass. β -particles can be stopped by about 1 cm thick sheet of aluminium or 1 m of air.

GAMMA RAYS

(1) **Nature.** Unlike α and β -rays, they do not consist of particles of matter. **γ -Rays are a form of electromagnetic radiation** of shorter wavelength than X-rays. They could be thought of as high-energy photons released by the nucleus during α - or β -emissions. They have no mass or charge and may be symbolized as ${}^0_0\gamma$.

(2) **Velocity.** Like all forms of electromagnetic radiation, **γ -rays travel with the velocity of light**.

(3) **Ionising power.** Their ionising power is very weak in comparison to α - and β -particles. A γ -photon displaces an electron of the gas molecule to yield a positive ion. Since the chances of photon-electron collisions are small, γ -rays are weak ionisers.

(4) **Penetrating power.** Because of their high velocity and non-material nature, **γ -rays are most penetrating**. They cannot be stopped even by a 5 cm thick sheet of lead or several metres thick layer of concrete.

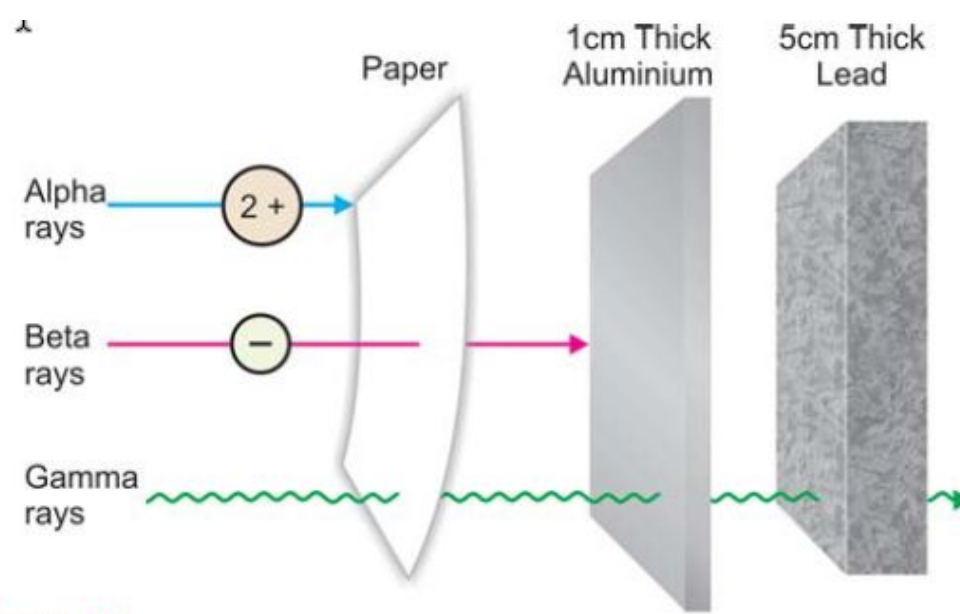


Figure 4.3
Penetrating powers of alpha, beta and gamma rays.

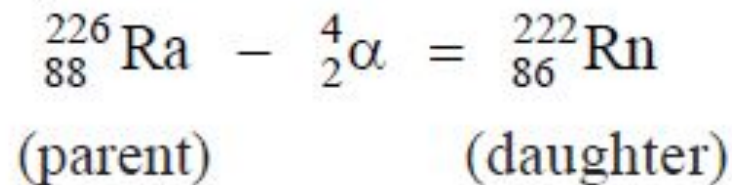
TABLE 4.1. COMPARISON OF PROPERTIES OF α , β AND γ -RAYS			
Property	α -rays	β -rays	γ -rays
Nature	helium nuclei ${}^4_2\text{He}$	fast electrons ${}^0_{-1}\text{e}$	electromagnetic radiation
Velocity	one-tenth of velocity of light	velocity of light	velocity of light
Penetrating Power	low	moderate	high
Stopped by	paper or 0.01 mm thick aluminium sheet	1 cm of aluminium	several cm thick lead/ concrete layer

RADIOACTIVE DECAY

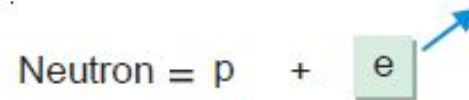
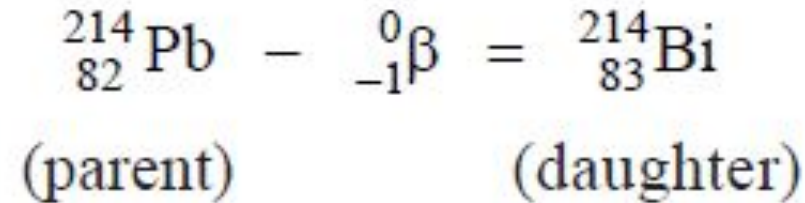
According to the theory put forward by Rutherford and Soddy (1903), radioactivity is a nuclear property. The nucleus of a radioactive atom is unstable. It undergoes decay or disintegration by spontaneous emission of an α - or β -particle. This results in the change of proton-neutron composition of the nucleus to form a more stable nucleus. **The original nucleus is called the parent nucleus and the product is called the daughter nucleus.**

Thus an α -emission reduces the atomic mass by 4 and atomic number by 2.

For example, Radium decays by α -emission to form a new element Radon,



Thus a β -emission increases the atomic number by 1 with no change in atomic mass.
An example of β -decay is the conversion of lead-214 to bismuth-214,

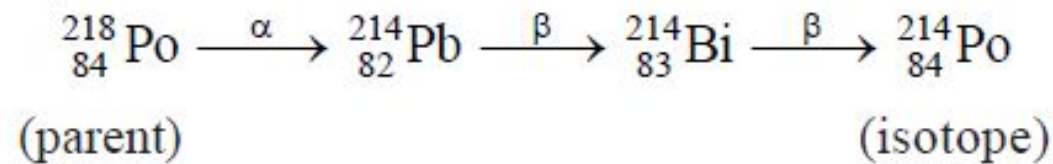


This results in the increase of one positive charge on the nucleus. The loss of a β -particle from the nucleus does not alter its atomic mass. For a parent nucleus with atomic mass M and atomic number Z , the daughter nucleus will have

$$\begin{aligned} \text{atomic mass} &= M \\ \text{atomic number} &= Z + 1 \end{aligned}$$

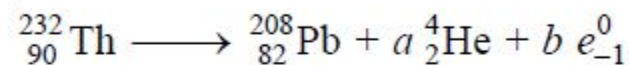
It is noteworthy that a β -emission results in the production of an isobar. Thus, $^{214}_{82}\text{Pb}$ and $^{214}_{83}\text{Bi}$ are isobaric as they have the same mass number 214 but different atomic numbers (82 and 83).

One α -emission and two β -emissions yield an isotope. Let us consider the following series of changes.



SOLVED PROBLEM 1. How many α and β particles are emitted in passing down from ${}_{90}^{232}\text{Th}$ to ${}_{82}^{208}\text{Pb}$.

SOLUTION. Let a be the number of α particles and b be the number of β particles emitted during the radioactive transformation. It can be represented as



Comparing the mass numbers, we get

$$232 = 208 + 4 \times a + b \times 0$$

or

$$4a = 232 - 208 = 24$$

or

$$a = 6$$

Comparing the atomic numbers, we get

$$90 = 82 + 2 \times a + b(-1)$$

Substituting the value of a , we get

$$90 = 82 + 2 \times 6 - b$$

or

$$b = 94 - 90$$

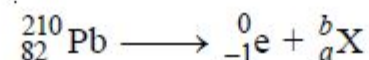
or

$$b = 4$$

Thus the number of α particles emitted = 6 and the number of β particles emitted = 4

SOLVED PROBLEM 2. $^{210}_{82}\text{Pb}$ is a β -emitter and $^{226}_{88}\text{Ra}$ is an α -emitter. What will be the atomic masses and atomic numbers of daughter elements of these radioactive elements ? Predict the position of daughter elements in the periodic table.

SOLUTION. (a) $^{210}_{82}\text{Pb}$ undergoes β -decay *i.e.*



Comparing the atomic masses, we have

$$210 = 0 + b$$

or

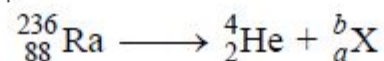
$$b = 210$$

and comparing the atomic numbers, we get

$$82 = -1 + a \text{ or } a = 83$$

Thus the daughter element will have the same atomic mass **210** and its atomic number will be **83**. It will occupy **one position right** to the parent element.

(b) $^{236}_{88}\text{Ra}$ undergoes α decay *i.e.*



Comparing the atomic masses, we get

$$236 = 4 + b \text{ or } b = 232$$

and comparing the atomic number, we get

$$88 = 2 + a \text{ or } a = 86$$

Thus the daughter element will have atomic mass **232** and its atomic number will be **86**. It will occupy **two positions to the left** of the parent element.

RADIOACTIVE DISINTEGRATION SERIES

A radioactive element disintegrates by the emission of an α - or β - particle from the nucleus to form a new '*daughter element*'. This again disintegrates to give another '*daughter element*'. The process of disintegration and formation of a new element continues till a non-radioactive stable element is the product.

The whole series of elements starting with the parent radioactive element to the stable end-product is called a Radioactive Disintegration Series.

Sometime, it is referred to as a **Radioactive Decay Series** or simply **Radioactive Series**. All the natural radioactive elements belong to one of the three series :

- (1) The Uranium Series
- (2) The Thorium Series
- (3) The Actinium Series

The Uranium Series

It commences with the parent element uranium-238 and terminates with the stable element lead-206. It derives its name from uranium-238 which is the prominent member of the series and has the longest half-life. The Uranium series is illustrated in Fig. 4.9.

The Thorium Series

It begins with the parent element thorium-232 and ends with lead-208 which is stable. This series gets its name from the prominent member thorium-232.

The Actinium Series

It starts with the radioactive element uranium-235. The end-product is the stable element lead-207. This series derives its name from the prominent member actinium-227.

THE URANIUM DISINTEGRATION SERIES

NUCLIDE	PARTICLE PRODUCED	HALF-LIFE
Uranium-238 ($^{238}_{92}\text{U}$)	α	4.51×10^9 years
↓		
Thorium-234 ($^{234}_{90}\text{Th}$)	β	24.1 days
↓		
Protactinium-234 ($^{234}_{91}\text{Pa}$)	β	6.75 hours
↓		
Uranium-234 ($^{234}_{92}\text{U}$)	α	2.48×10^5 years
↓		
Thorium-230 ($^{230}_{90}\text{Th}$)	α	8.0×10^4 years
↓		
Radium-226 ($^{226}_{88}\text{Ra}$)	α	1.62×10^3 years
↓		
Radon-222 ($^{222}_{86}\text{Rn}$)	α	3.82 days

Polonium-218 ($^{218}_{84}\text{Po}$)	α	3.1 minutes
↓		
Lead-214 ($^{214}_{82}\text{Pb}$)	β	26.8 minutes
↓		
Bismuth-214 ($^{214}_{83}\text{Bi}$)	β	19.7 minutes
↓		
Polonium-214 ($^{214}_{84}\text{Po}$)	α	1.6×10^{-4} second
↓		
Lead-210 ($^{210}_{82}\text{Pb}$)	β	20.4 years
↓		
Bismuth-210 ($^{210}_{83}\text{Bi}$)	β	5.0 days
↓		
Polonium-210 ($^{210}_{84}\text{Po}$)	α	138.4 days
↓		
Lead-206 ($^{206}_{82}\text{Pb}$)	—	Stable

RATE OF RADIOACTIVE DECAY

The decay of a radioactive isotope takes place by disintegration of the atomic nucleus. It is not influenced by any external conditions. Therefore the rate of decay is characteristic of an isotope and depends only on the number of atoms present. If N be the number of undecayed atoms of an isotope present in a sample of the isotope, at time t ,

$$-\frac{dN}{dt} \propto N$$

or

$$-\frac{dN}{dt} = \lambda N \quad \dots (1)$$

where $-\frac{dN}{dt}$ means the rate of decrease in the number of radioactive atoms in the sample; and λ is the proportionality factor. This is known as the **decay constant** or **disintegration constant**. Putting $dt = 1$ in equation (1) we have

$$-\frac{dN}{N} = \lambda \quad \dots (2)$$

Thus decay constant may be defined as the proportion of atoms of an isotope decaying per second.

dps: disintegration per second

UNITS OF RADIOACTIVITY

The standard unit of radioactivity (*i.e.* rate of disintegration) is *Curie* (c). A curie is a quantity of radioactive material decaying at the same rate as 1 g of Radium (3.7×10^{10} dps). Rutherford is a more recent unit.

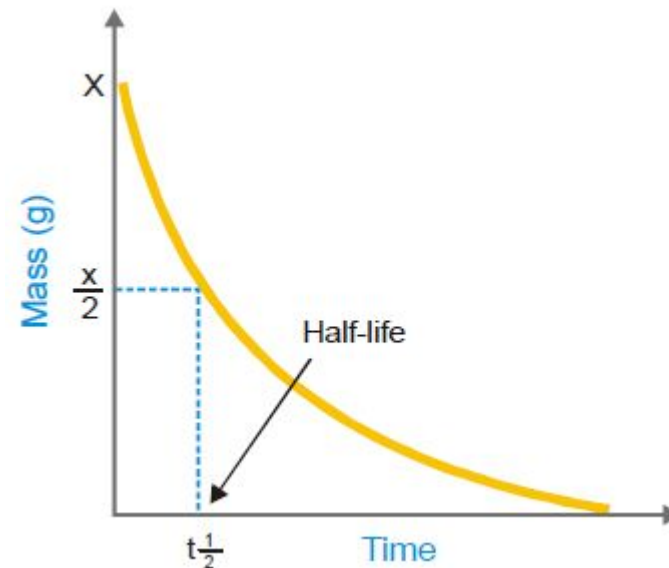
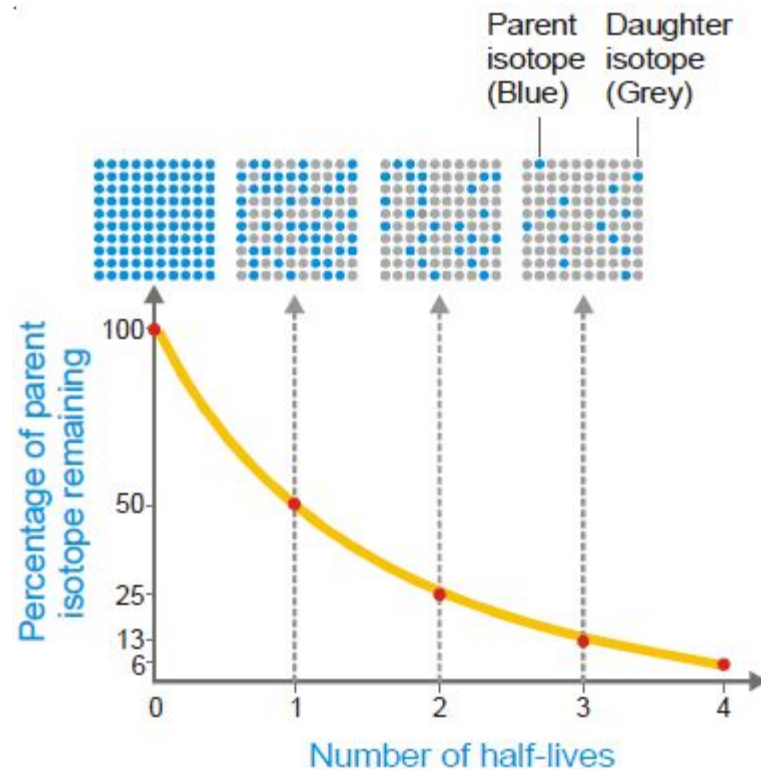
$$1 \text{ Rutherford} = 10^6 \text{ dps}$$

The S.I. unit is Becquerel

$$1 \text{ Bq} = 1 \text{ dps}$$

HALF-LIFE

The half-life or half-life period of a radioactive isotope is the time required for one-half of the isotope to decay. Or, it may be defined as the time for the radioactivity of an isotope to be reduced to **half of its original value**. Half-life period is characteristic of a radioactive element. For example, the half-life of radium is 1620 years. This means that 1 g of radium will be reduced to 0.5 g in 1620 years and to 0.25 g in further 1620 years; and so on. Some other radioactive elements may have half-life of a fraction of a second and for others it may be millions of years. The unit of half-life period is time^{-1} .



From equation (1) we can write

$$-\frac{dN}{N} = \lambda dt$$

On integration,

$$-\int \frac{dN}{N} = \lambda \int dt$$

or $-\ln N = \lambda t + X \text{ (constant)}$... (3)

If N_0 is the number of atoms at time $t = 0$, $X = -\ln N_0$

Substituting the value of X in (3)

$$-\ln N = \lambda t - \ln N_0$$

or $\ln \left(\frac{N_0}{N} \right) = \lambda t$

Using ordinary logs,

$$2.303 \log \left(\frac{N_0}{N} \right) = \lambda t$$
 ... (4)

At half-life time ($t_{1/2}$), $N = 1/2 N_0$

$$\therefore 2.303 \log \left(\frac{N_0}{\frac{1}{2}N_0} \right) = 2.303 \log 2 = \lambda t_{1/2}$$

or $0.693 = \lambda t_{1/2}$

or $t_{1/2} = \frac{0.693}{\lambda}$... (5)

The value of λ can be found experimentally by finding the number of disintegrations per second with the help of a Geiger-Muller counter. Hence, half-life of the isotope concerned can be calculated by using the relation (5).

Calculation of Half-life of a First order Reaction

The integrated rate equation (4) for a first order reaction can be stated as :

$$k = \frac{2.303}{t} \log \frac{[A]_0}{[A]}$$

where $[A]_0$ is initial concentration and $[A]$ is concentration at any time t . Half-life, $t_{1/2}$, initial concentration reduces to $\frac{1}{2}$ i.e.,

$$[A] = \frac{1}{2}[A]_0$$

Substituting values in the integrated rate equation, we have

$$k = \frac{2.303}{t_{1/2}} \log \frac{[A]_0}{1/2[A]_0} = \frac{2.303}{t_{1/2}} \log 2$$

or $t_{1/2} = \frac{2.303}{k} \log 2 = \frac{2.303}{k} \times 0.3010$

or $t_{1/2} = \frac{0.693}{k}$

SOLVED PROBLEM 1. Calculate the half-life of radium-226 if 1 g of it emits 3.7×10^{10} alpha particles per second.

SOLUTION

Rate of decay = Rate of emission of α -particles

We know that

$$\frac{dN}{dt} = \lambda N = 3.7 \times 10^{10} \text{ per second}$$

The number of atoms of radium present (N) in 1g of sample, $= \frac{6.023 \times 10^{23}}{226}$

From equation (5) stated earlier $\lambda = \frac{0.693}{t_{1/2}}$

Substituting the value of λ and N in equation (1) above

$$\frac{dN}{dt} = \frac{0.693}{t_{1/2}} \times \frac{6.023 \times 10^{23}}{226} = 3.7 \times 10^{10}$$

Hence

$$\begin{aligned} t_{1/2} &= \frac{0.693 \times 6.023 \times 10^{23}}{3.7 \times 10^{10} \times 226 \times 60 \times 60 \times 24 \times 365} \\ &= 1583 \text{ years} \end{aligned}$$

⚡ **SOLVED PROBLEM 2.** Calculate the disintegration constant of cobalt 60 if its half-life to produce nickel – 60 is 5.2 years.

SOLUTION

From equation (5) stated earlier $t_{1/2} = \frac{0.693}{\lambda}$

or
$$\lambda = \frac{0.693}{t_{1/2}}$$

Substituting the value of $t_{1/2}$, we have $\lambda = \frac{0.693}{5.2 \text{ yr}} = 0.13 \text{ yr}^{-1}$

SOLVED PROBLEM 3. The half-life period of radon is 3.825 days. Calculate the activity of radon. (atomic weight of radon = 222)

SOLUTION

We know that

$$dN = \lambda N \quad \dots (2)$$

where dN is the number of atoms disintegrating per second, λ is the decay constant and N is the number of atoms in the sample of radon.

Calculation of λ :

$$\lambda = \frac{0.693}{t_{1/2}} = \frac{0.693}{3.825 \times 24 \times 60 \times 60} = 2.096 \times 10^{-6} \text{ sec}^{-1}$$

Calculation of N :

From expression (1) above

$$N = \frac{dN}{\lambda} = \frac{3.7 \times 10^{10}}{2.096 \times 10^{-6}} = 1.7653 \times 10^{16} \text{ atoms}$$

Mass of 6.02×10^{23} atoms of radon = 222 g

$$\begin{aligned} \text{Mass of } 1.7653 \times 10^{16} \text{ atoms of radon} &= \frac{222}{6.02 \times 10^{23}} \times 1.7653 \times 10^{16} \\ &= 6.51 \times 10^{-6} \text{ g} \end{aligned}$$

By definition, the activity of radon is its mass in grams which gives 3.7×10^{10} disintegrations per second. Therefore **activity of radon = 6.51×10^{-6} g curie.**

CALCULATION OF SAMPLE LEFT AFTER TIME t

It follows from equation (4) stated earlier that

$$\log\left(\frac{N_0}{N}\right) = \frac{\lambda t}{2.303}$$

Knowing the value of λ , the ratio of N_0/N can be calculated. If the amount of the sample present to start with is given, the amount left after lapse of time t can be calculated.

SOLVED PROBLEM 1. Cobalt-60 disintegrates to give nickel-60. Calculate the fraction and the percentage of the sample that remains after 15 years. The disintegration constant of cobalt-60 is 0.13 yr^{-1} .

SOLUTION

$$\log\frac{N_0}{N} = \frac{\lambda t}{2.303} = \frac{(0.13 \text{ yr}^{-1})(15 \text{ yr})}{2.303} = 0.847$$

$$\frac{N_0}{N} = \text{antilog } 0.847 = 7.031$$

The fraction remaining is the amount at time t divided by the initial amount.

$$\frac{N}{N_0} = \frac{1}{7.031} = 0.14$$

Hence the fraction remaining after 15 years is 0.14 or 14 per cent of that present originally.

SOLVED PROBLEM 2. How much time would it take for a sample of cobalt-60 to disintegrate to the extent that only 2.0 per cent remains ? The disintegration constant λ is 0.13 yr^{-1} .

SOLUTION

$$\frac{N}{N_0} = \frac{2}{100} = 0.02$$

or

$$\frac{N_0}{N} = \frac{1}{0.02} = 50$$

From equation (4) stated earlier

$$\log \left(\frac{N_0}{N} \right) = \frac{\lambda t}{2.303}$$

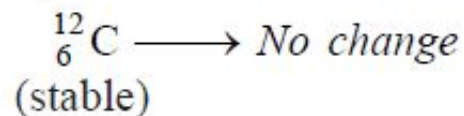
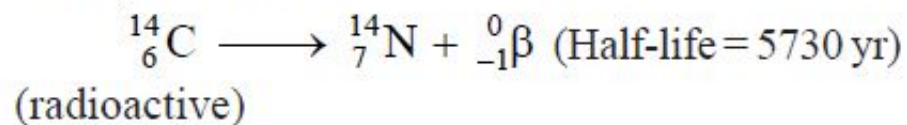
$$\log 50 = \frac{(0.13 \text{ yr}^{-1}) t}{2.303}$$

or

$$t = \frac{2.303 \log 50}{0.13 \text{ yr}^{-1}} = 30 \text{ years}$$

RADIOACTIVE DATING

The age of an old piece of wood can be determined by radioactive dating technique. The atmosphere contains radioactive carbon dioxide, $^{14}\text{CO}_2$, and ordinary carbon dioxide, $^{12}\text{CO}_2$, in a fixed ratio. A plant while alive takes up both types of carbon dioxide and converts them to carbon-14 and carbon-12 photosynthesis. Thus a living plant contains radioactive carbon-14 and stable carbon-12 in a fixed ratio. When the plant dies, the uptake of carbon from the atmosphere stops. Now onward, carbon-12 remains unchanged but carbon-14 decays by beta-emission.



As a result, $^{14}\text{C}/^{12}\text{C}$ decreases with lapse of time.

Therefore the concentration of carbon-14 declines with time. The concentration of carbon-14 can be measured by counting radioactivity. Knowing the concentration of carbon-14 in a given sample of old wood and that in a living plant, the age of the sample can be calculated.

SOLVED PROBLEM. The amount of carbon-14 in a piece of wood is found to be one-sixth of its amount in a fresh piece of wood. Calculate the age of old piece of wood.

SOLUTION

Calculation of Decay constant :

The half-life of carbon-14 = 5730 years

$$\therefore \lambda = \frac{0.693}{t_{1/2}} = \frac{0.693}{5730 \text{ yr}}$$

Calculation of Age of Wood :

From expression (4) on page 89

$$t = \log \frac{{}^{14}\text{C}_0}{{}^{14}\text{C}_t} \times \frac{2.303}{\lambda}$$

or

$$\begin{aligned} t &= \log \frac{1}{1/6} \times \frac{2.303 \times 5730}{0.693} \\ &= \log 6 \times \frac{2.303 \times 5730}{0.693} \\ &= \frac{0.7782 \times 2.303 \times 5730}{0.693} \\ &= \mathbf{14818.5 \text{ years}} \end{aligned}$$

Changing into common logarithms

$$k = \frac{2.303}{t} \log \frac{a}{a-x}$$

$$k = \frac{2.303}{t} \log \frac{[\text{A}]_0}{[\text{A}]}$$

SOLVED PROBLEM. A bone taken from a garbage pile buried under a hill-side had $^{14}\text{C}/^{12}\text{C}$ ratio 0.477 times the ratio in a living plant or animal. What was the date when the animal was buried?

SOLUTION

Half-life of carbon-14 is 5730 years

$$\therefore \lambda = \frac{0.693}{t_{1/2}} = \frac{0.693}{5730 \text{ yr}}$$

Substituting values in the expression

$$\begin{aligned} t &= \log \frac{{}^{14}\text{C}_0}{{}^{14}\text{C}_t} \times \frac{2.303}{\lambda} \\ &= \log \frac{1.0}{0.477} \times \frac{2.303 \times 5730}{0.693} = 6.1 \times 10^3 \\ &= \mathbf{6100 \text{ yr}} \end{aligned}$$

The animal was buried 6100 years ago.

NUCLEAR REACTIONS

In a chemical reaction there is merely a rearrangement of extranuclear electrons. The atomic nucleus remains intact. A nuclear reaction involves a change in the composition of the nucleus. The number of protons and neutrons in the nucleus is altered. The product is a new nucleus of another atom with a different atomic number and/or mass number. Thus,

A nuclear reaction is one which proceeds with a change in the composition of the nucleus so as to produce an atom of a new element.

The conversion of one element to another by a nuclear change is called **transmutation**.

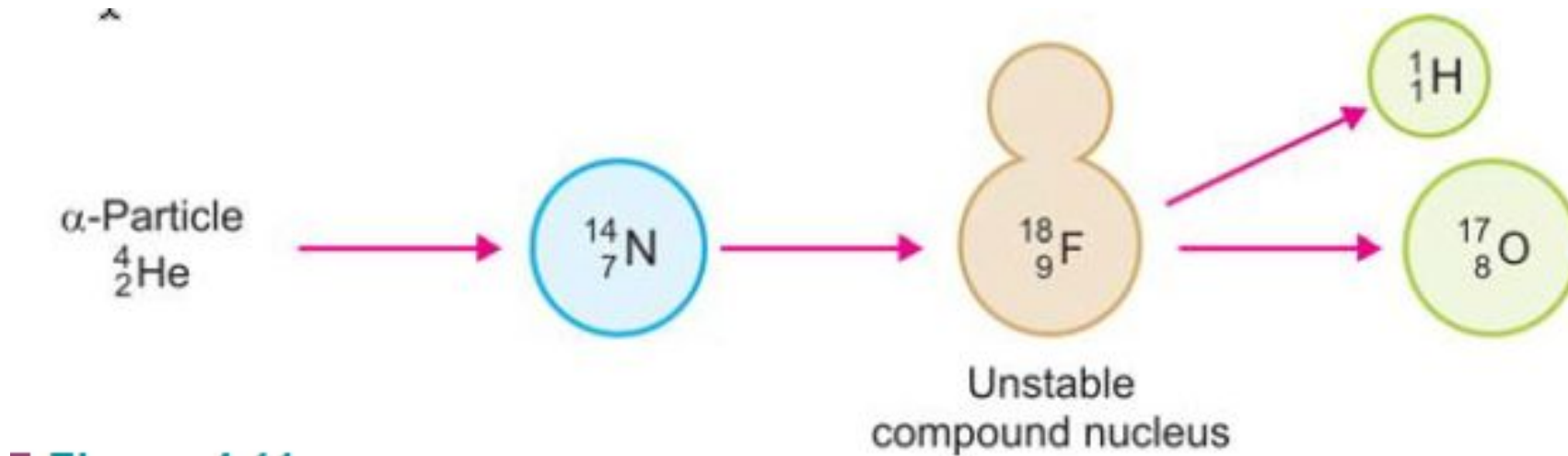
We have already considered the nuclear reactions of radioactive nuclei, producing new isotopes. Here we will consider such reactions caused by artificial means.

DIFFERENCES BETWEEN NUCLEAR REACTIONS AND CHEMICAL REACTIONS

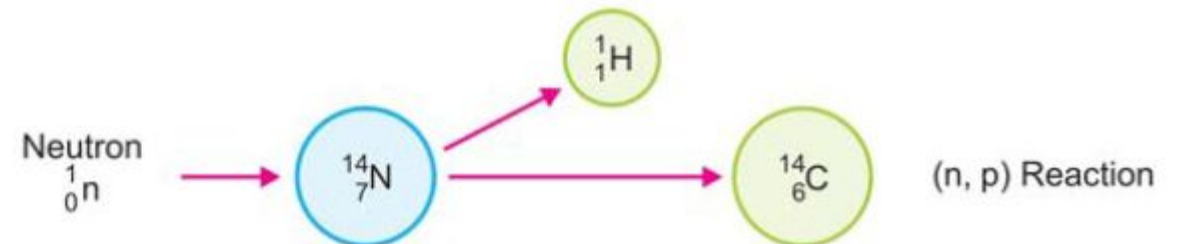
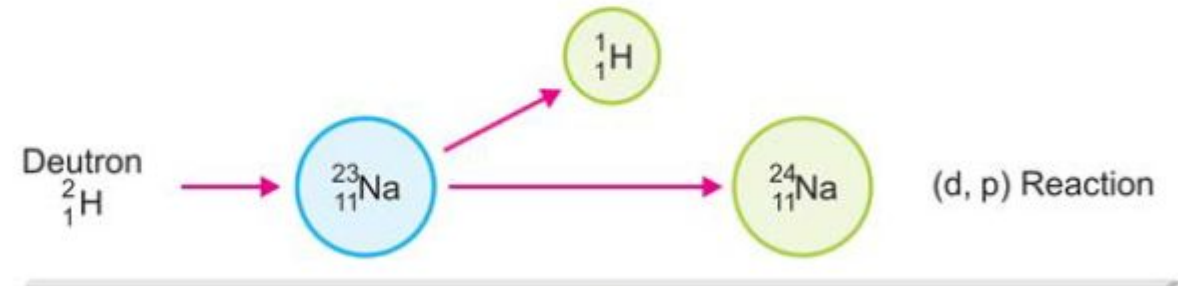
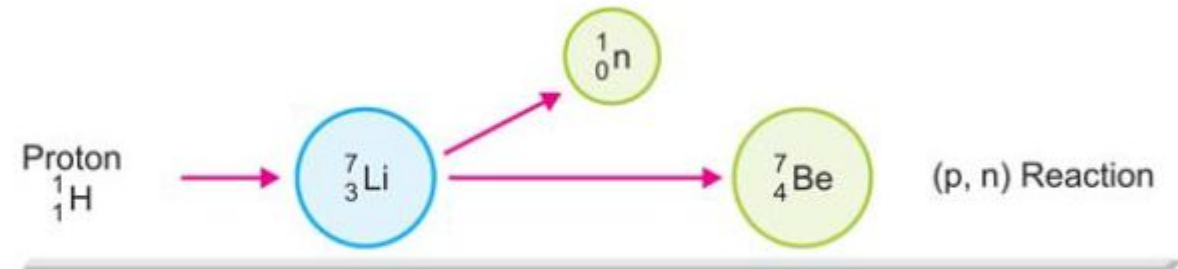
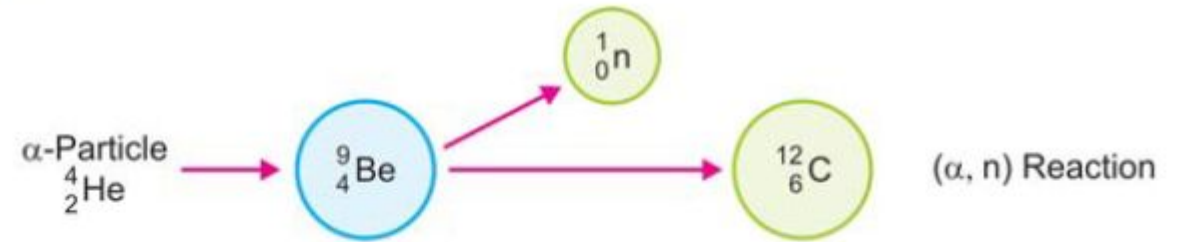
Nuclear Reactions	Chemical Reactions
1. Proceed by redistribution of nuclear particles. 2. One element may be converted into another. 3. Often accompanied by release or absorption of enormous amount of energy. 4. Rate of reaction is unaffected by external factors such as concentration, temperature, pressure and catalyst.	1. Proceed by the rearrangement of extra-nuclear electrons. 2. No new element can be produced. 3. Accompanied by release or absorption of relatively small amount of energy. 4. Rate of reaction is influenced by external factors.

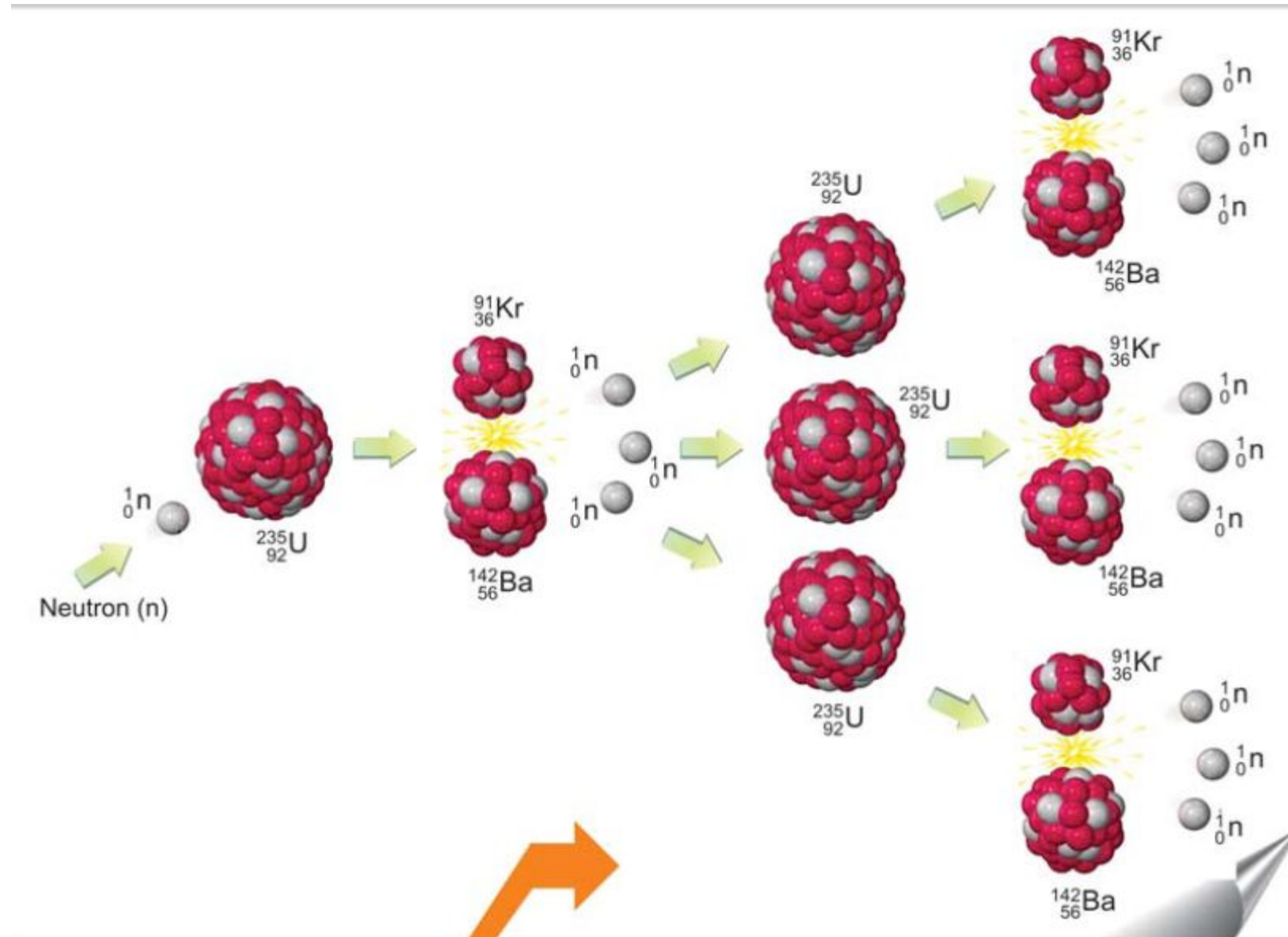
NUCLEAR FISSION REACTIONS

In these reactions an atomic nucleus is broken or fissioned into two or more fragments. This is accomplished by bombarding an atom by alpha particles (${}^4_2\text{He}$), neutrons (${}_0^1n$), protons (${}_1^1\text{H}$), deuterons (${}_1^2\text{H}$), etc.



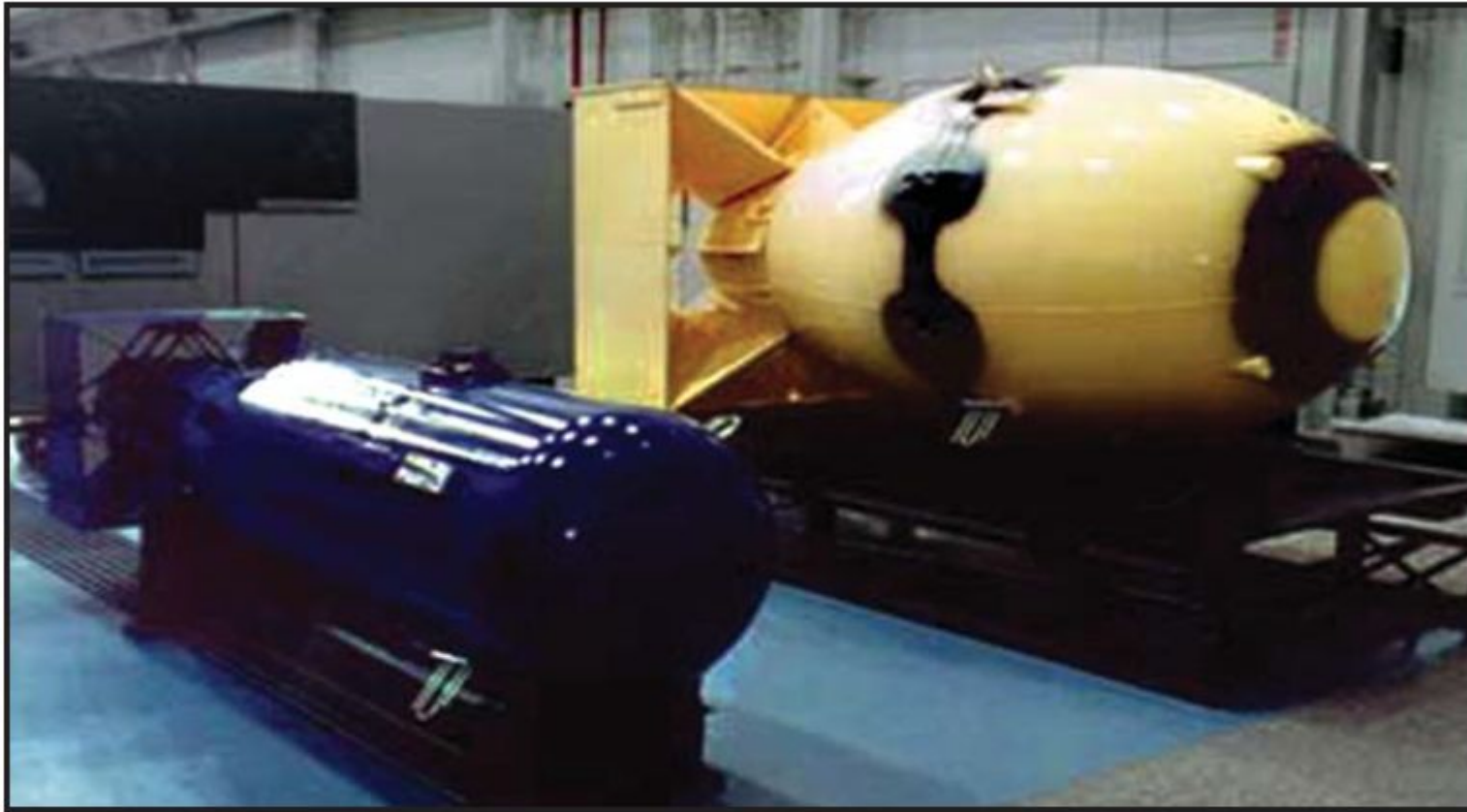
Other examples are,





THE ATOMIC BOMB

A bomb which works on the principle of a fast nuclear chain reaction is referred to as the **atomic bomb**. A design of such a bomb is shown in Fig. 4.19. It contains two subcritical masses of fissionable material, ^{235}U or ^{239}Pu . It has a mass of trinitrotoluene in a separate pocket. When TNT is detonated, it drives one mass of ^{235}U into the other. A supercritical mass of the fissionable material is obtained. As a result of the instantaneous chain reaction, the bomb explodes with the release of tremendous heat energy. **Temperature developed in an atomic bomb is believed to be 10 million $^{\circ}\text{C}$ (temperature of the sun)**. Besides many radionuclei and heat, deadly gamma rays are released.



■ The First Atomic Bombs

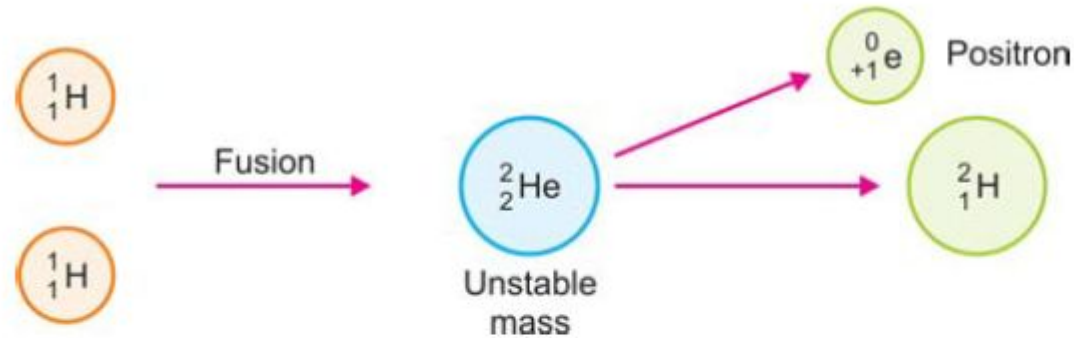
Little Boy was the first nuclear weapon used in warfare. It exploded approximately 1,800 feet over Hiroshima, Japan, on the morning of August 6, 1945, with a force equal to 13,000 tons of TNT. Immediate deaths were between 70,000 to 130,000.

Fat Man was the second nuclear weapon used in warfare. Dropped on Nagasaki, Japan, on August 9, 1945, *Fat Man* devastated more than two square miles of the city and caused approximately 45,000 immediate deaths.

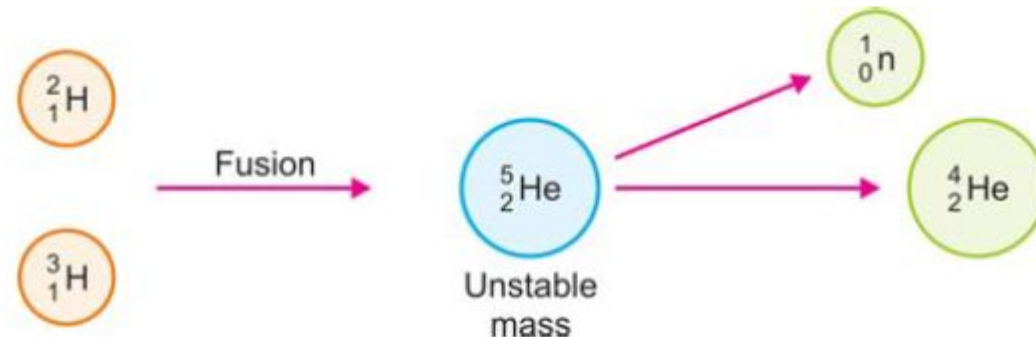
NUCLEAR FUSION REACTIONS

These reactions take place by combination or fusion of two small nuclei into a larger nucleus. At extremely high temperatures the kinetic energy of these nuclei overweighs the electrical repulsions between them.

- (1) Two hydrogen nuclei, ${}^1_1\text{H}$, fuse to produce a deuterium nucleus, ${}^2_1\text{H}$.



- (2) Deuterium nucleus, ${}^2_1\text{H}$, and tritium nucleus, ${}^3_1\text{H}$, combine to give helium nucleus, ${}^4_2\text{He}$ with the expulsion of a neutron.



Nuclear Fission

1. A bigger (heavier nucleus splits into smaller (lighter) nuclei.
2. It does not require high temperature.
3. A chain reaction sets in.
4. It can be controlled and energy released can be used for peaceful purposes.
5. The products of the reaction are radioactive in nature.
6. At the end of the reaction nuclear waste is left behind.

Nuclear Fusion

1. Lighter nuclei fuse together to form the heavier nucleus.
2. Extremely high temperature is required for fusion to take place.
3. It is not a chain reaction.
4. It cannot be controlled and energy released cannot be used properly.
5. The products of a fusion reaction are non-radioactive in nature.
6. No nuclear waste is left at the end of fusion reaction.

Disposal of waste

The products of fission *e.g.*, Ba-139 and Kr-92, are themselves radioactive. They emit dangerous radiation for several hundred years. The waste is packed in concrete barrels which are buried deep in the earth or dumped in the sea. But the fear is that any leakage and corrosion of the storage vessels may eventually contaminate the water supplies.